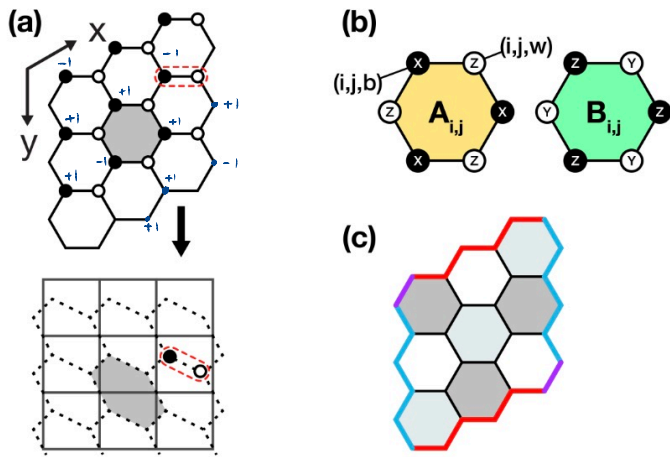




Assume in x basis.

- A - stabilizer constrains only 2 qubits are in $+1$.

The 3-element sublattice $\Rightarrow$ 3-cell basis $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$

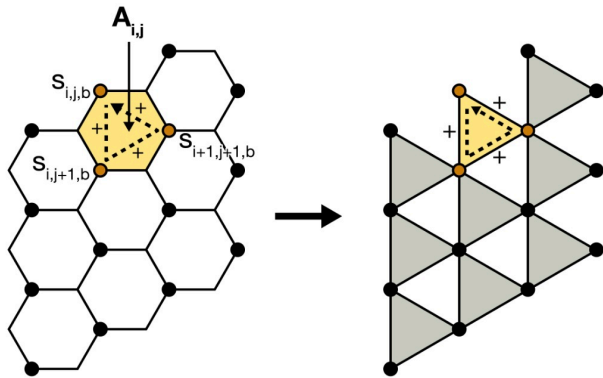
stabilizer condition $(+1)$ gives the update rule

- white sublattice is in equal - superposition

$$[A_{ij}, B_{ij}] = 0 \Rightarrow \text{simultaneous eigenstate}$$

$$A_{ij} = -\hat{X} \hat{Z} \hat{X} \hat{Z} \hat{X} \hat{Z} \dots$$

$$B_{ij} = U_1^\dagger \hat{X} U_1 U_1^\dagger \hat{Z} U_1 U_1^\dagger \hat{X} U_1 U_1^\dagger \hat{Z} U_1 U_1^\dagger \hat{X} U_1 U_1^\dagger \hat{Z} U_1 U_1^\dagger \hat{X} U_1$$



$$H_{NM} = \sum_{ij, k \in \Delta} (\sigma_i + \sigma_j + \sigma_k \mod 2)$$

since opposite spins have lower energy

$$\begin{array}{ll} 1000 & 1111 \\ 1001 & 1011 \end{array}$$

conventions: $s_i = 0 \uparrow$
 $s_i = 1 \downarrow$

At infinitely biased sublattice noise, not correlated

$$H_{AB} = - \sum_{ij} A_{ij} + B_{ij} \quad \geq \text{disjoint copies.}$$

At finitely biased noise, there is permutation symmetry under U_q .

$$H = \sum_{ij} A_{ij} + B_{ij} + A_{ij} B_{ij}$$

Probabilistic Cellular Automaton.

states $\{s^1, s^2, \dots, s^n\}$, over time s^t, s^T

Transitions P_{ij} (n/n)

local transition $s_k^t = s_k^j$ if $k \in B_r(q)$.

$$P_{ij} = f(s_k^i, s_k^j \mid k \in B_r(q))$$

w the change in energy.

$|0\rangle \rightarrow |1\rangle$ w could be ± 1 for a single cell.

$$G_q(w) = e^{\beta w} G_q(-w) \quad \text{like a symmetric constraint.}$$

detailed balance equation, in equilibrium with reversible process.

$$G(w) = \frac{w}{1 - e^{\beta w}}$$

$$\gamma_q(w) = \frac{w}{1 - e^{\beta w}} - \gamma_z$$

At equilibrium of automaton, $\gamma_q(w) = 0$.

$$\gamma_z = \frac{w}{1 - e^{\beta w}}$$

$$\beta = \frac{1}{\gamma} \ln \left(\frac{6 + \gamma_z}{\gamma_z} \right)$$

$$\Rightarrow T_{\max} [d, \gamma_z]$$